Dae Seok Song

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AI Engineer | Medical AI and Modular Neural Networks

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Research Profile

AI engineer who has led model development, compression, clinical-product integration, and international regulatory evidence generation at a medical-AI startup since November 2022. My research asks whether a functional module keeps its role when moved into a separately trained model (so far tested only across models that differ only in initialization), and tests this with intervention: swapping a module between separately trained models and measuring what breaks (preliminary; removal is used only as a control in a separate setup).

Research Interests

Functional-module transplant and intervention; representation alignment; modular and continual learning; model compression and structured pruning; local learning rules.

Independent Research

RevolverNet: Detachable Functional Modules and Conditional Routing 2026–Present

Independent Researcher Seoul, Korea

- Designed a modular architecture with interchangeable slots, detachable functional modules, and a router.
- Across two completed image-classification conditions, one-time representation alignment recovered 99.4–99.8% of the performance lost under naive swaps, covering every directed donor-to-recipient pair among four separately trained models. The models shared the dataset, architecture, and role assignment (roles were assigned by me, not learned) but differed in initialization. Preliminary, unpublished independent work; prior work on model stitching and re-basin already aligns independently trained networks, so the recovery itself is not claimed as new.
- In a separate setup that loads multiple modules together without the router (not the four transplant models above), measured the residual task-performance loss after removing a single module across designs matched on accuracy (0.03–0.77; preliminary, single seed, noise floor $\pm$0.019). I use removal only as a control, not as standalone evidence that a module is load-bearing.
- Implemented routing so that a router selects one module for each input.
- Long-term interest: module-level control of AI behavior, i.e., removing parts that could harm people. Prior work on module-level intervention exists (e.g., Gradient Routing, SteerMoE); defining and separating harm is an open problem, removal appears here only as a preliminary control and the removal comparison is planned, and current results cover only module compatibility.

Brain-Inspired vs. Backpropagation Learning-Rule Comparison 2026

Independent Researcher Seoul, Korea

- Compared a burst-plasticity-style local learning rule against backpropagation on the same modular architecture, implemented at the firing-rate level (did not reach a spiking implementation).
- Backpropagation led on both accuracy (three-seed average 90.8% vs. 77.5%) and post-transplant module performance; the result redirected the project from “which rule learns better” toward intervention-based measurement of what a learned module actually preserves.

Professional Experience

Connecteve Co., Ltd. Nov 2022–Present

AI Engineer / Early Team Member Seoul, Korea

- Consolidated 10 production knee X-ray models into 2 uncertainty-aware, partial-label multi-task models and added a keypoint model (3 models in total). Against the current product engine on the same CPU server and bilateral films, stored weights are 10.97$\times$ smaller (multi-task consolidation plus a tiny backbone) and wall-clock inference time per image is 1.74$\times$ shorter; the overall F1 on evaluation sets from three external institutions did not drop (0.016 higher; single seed).
- Rebuilt the optimization-to-serving line (ONNX optimization, 16-bit inference, BentoML serving): engine size 47% smaller and inference 3$\times$ faster than the same engine before optimization, with clinical validation showing only a 0.06-percentage-point accuracy drop and 2.1% grade changes (non-inferiority confirmed by patient-level bootstrap).

- Tuned per-grade decision thresholds with Youden's J statistic across four tie-break strategies, surpassing an FDA-cleared competitor product's F1 on a matched evaluation.
- Built the standard PyTorch Lightning + Hydra experiment pipeline for medical foundation models (DINOv3, Rad-DINO) with selectable linear probing or LoRA.
- Built an in-house MLOps console (GUI/CLI) with a two-stage hyperparameter search (LLM-agent proposals, then Optuna/TPE on plateau) that raised detection mAP from 0.80 to 0.87; evaluated 12 agentic-workflow frameworks and selected LangGraph, running agents on vLLM (Qwen, Mistral); caching cut re-onboarding from 3 h 36 min to 38 min (5.7×).
- Led AI technical documentation for medical-device approval in Korea (MFDS), the EU (CE MDR), the UAE (DOH), and Singapore (HSA).

Minerva Soft Mar 2022–Jul 2022
Image Processing Intern Korea

- Replaced full-image barcode scanning with an OpenCV ROI pipeline, raising recognition from 70% to 90%; shipped a C++ DLL after 170,000+ memory-leak and stability iterations.

Education

University of Suwon Mar 2017–Feb 2023
B.S. in Information Security, Valedictorian GPA 4.29/4.50; Major 4.37/4.50

Selected Publications

Lee, D. W., **Song, D. S.**, et al. "Accurate, automated classification of radiographic knee osteoarthritis severity using a novel method of deep learning: Plug-in modules." *Knee Surgery & Related Research*, 2024. Second author; devised and implemented the core plug-in-module ensemble that imitates clinicians' consensus, with the paper reporting an accuracy improvement of about 6%. doi:10.1186/s43019-024-00228-3

"Image size transformation data augmentation method for lesion segmentation on normalization and augmentation medical images." *Korea Software Congress (KSC)*, 2022, poster presentation. First author.

Patents & Other Research Output

- Inventor, Korean Patent Application 10-2024-0124529 and PCT/KR2025/012689: ensemble system for knee-osteoarthritis severity classification (the plug-in-module method above).
- Inventor, Korean Patent Application 10-2025-0043898: generating a patient's projected knee image from the current radiograph and its K–L/OARSI grades.

Technical Strengths

Research	Controlled ablation design, multi-seed evaluation, intervention-based functional analysis, representation analysis, patient-level bootstrap non-inferiority tests, reproducibility
ML	PyTorch, PyTorch Lightning, Hydra, multi-task and partial-label learning, foundation-model adaptation (linear probing, LoRA), medical-image classification/segmentation
Compression & serving	ONNX, OpenVINO 16-bit, quantization and structured/block pruning, TorchServe, BentoML, FastAPI, vLLM
Engineering	Python, C/C++, Docker, GPU server operation (H100, L40S), regulated-product documentation

Languages

Korean (native); English (conversational); Japanese (conversational).